

Load Interconnection Request System Impact Study Report

For Queue Project Q22-2242

Clutch Switch 138 kV
January 03, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2242. The Interconnection Customer proposes a 202.6 MW large load (data center) facility interconnecting at the Clutch Switch 138 kV substation in the Virginia area. The request was received on November 29, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 707 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,372,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Belagio to Clutch Switch 138.0 kV	Axton Substation to Riverview 138.0 kV	107.8	242	100.0
Riverview Substation to Axton Substation 138.0 kV	Clutch Switch to Belagio 138.0 kV	101.0	258	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cltch Switch to Belagio Substation 138.0 kV	Reconductor line	\$3,264,000	100% (\$3,264,000)
Riverview Substation to Axton 138.0 kV	Reconductor line	\$2,108,000	100% (\$2,108,000)
	Total allocated to project		\$5,372,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.